Literature Review

HR Data Analysis

Literature Review

A literature review is a critical evaluation and synthesis of published research on HR analytics, not a summary of papers. It places this project within the wider academic framework.

Establishing Context and Foundational Knowledge

HR analytics, also called people analytics, applies data analysis to workforce data to support decisions on hiring, retention, performance, and compensation. It has moved from basic HR reporting to evidence-based management, where dashboards and statistical analysis explain why employees leave and what keeps them engaged. The main debates concern which factors drive attrition (pay, overtime, job satisfaction, tenure, work-life balance, career growth) and how to use employee data fairly and with respect for privacy.

Identifying the Research Gap

Much of the existing work stops at descriptive statistics, reporting attrition rates without showing which factors matter most or which employee groups are at risk. Other studies focus on a single factor, such as salary, and reach conflicting conclusions. Few turn their findings into an interactive tool that HR managers can use directly. This project addresses these gaps by combining multi-factor exploratory analysis with an interactive dashboard.

Justifying the Research and Methodology

Because earlier work is often descriptive, this project asks a more specific question: what drives employee attrition, and which segments are most affected? Answering it needs multi-factor analysis, so the methodology uses data cleaning, segmentation by department, job role, tenure, and income band, and correlation analysis, the techniques the literature relies on to find attrition drivers. A dashboard is used so that HR decision-makers can explore the results themselves.

Avoiding Duplication of Effort

Earlier studies have already linked factors such as overtime, low satisfaction, and low pay to attrition, so this project does not try to prove those links in general. Instead, it measures how strongly each factor relates to attrition in this dataset and turns the results into segment-level insights and recommendations.

Formulating a Theoretical or Conceptual Framework

The analysis groups the variables into five families and follows a fixed process.

- Demographic: age, gender, marital status.
- Job-related: department, job role, job level, tenure, overtime.
- Compensation: monthly income, salary hike, stock options.
- Satisfaction: job satisfaction, environment satisfaction, work-life balance.
- Outcome: attrition (stayed or left).

Process: data cleaning, exploratory analysis, KPI definition (attrition rate, average tenure, average income by group), segmentation and correlation analysis, then dashboard design.

Summary

The literature shows that HR analytics is an established practice, that attrition has known drivers, and that gaps remain in multi-factor segment analysis and actionable dashboards. This gives the project a solid basis and positions it as a practical contribution to HR decision-making.

Feedback & Evaluation

Lecturer's assessment of the project.

[To be completed by the lecturer.]

Suggested Improvements

Areas where the project can be enhanced.

- Add a predictive model (such as logistic regression or a decision tree) to estimate each employee's attrition risk.
- Use a larger or more recent dataset, ideally from several companies, so the findings generalize better.
- Automate data refresh in the dashboard so HR sees up-to-date figures without manual work.
- Add cost analysis, such as the estimated cost of replacing employees who leave, to link attrition to business impact.
- Include fairness checks across gender, age, and department to make sure the analysis does not reinforce bias.

Final Grading Criteria

Breakdown of marks based on documentation, implementation, testing, and presentation.

Criterion	Max Marks	Awarded
Documentation	[ ]	[ ]
Implementation	[ ]	[ ]
Testing	[ ]	[ ]
Presentation	[ ]	[ ]
Total	**[ ]**	**[ ]**